

Assignment 3 Report

Q.1

1. Objective

The objective of this study is to predict customer purchase behavior using supervised learning models and to evaluate the impact of class imbalance handling. Logistic Regression (with and without class balancing) and K-Nearest Neighbour (KNN) were implemented and compared using standard classification metrics.

2. Dataset Overview

The dataset contains customer demographic information including **Gender, Age, and Estimated Salary**, with a binary target variable **Purchased**. The data exhibits **moderate class imbalance**, with non-purchasers forming the majority class. This motivates the use of imbalance-aware modeling and evaluation beyond accuracy alone.

3. Logistic Regression (Without Class Balancing)

Test Set Performance:

- Accuracy: **0.81**
- Precision: **0.815**
- Recall: **0.611**
- F1-score: **0.698**
- MSE: **0.1197**
- RMSE: **0.3460**

Confusion Matrix:

$$\begin{bmatrix} 59 & 5 \\ 14 & 22 \end{bmatrix}$$

Analysis:

The unbalanced logistic regression model achieves reasonable accuracy and high precision, indicating that predicted buyers are mostly correct. However, recall is relatively low, meaning a significant portion of actual buyers are misclassified as non-buyers. This behavior reflects a conservative model biased toward the majority class.

4. Logistic Regression (With Class Balancing)

Test Set Performance:

- Accuracy: **0.83**
- Precision: **0.744**
- Recall: **0.806**
- F1-score: **0.773**
- MSE: **0.1225**

- RMSE: **0.3501**

Confusion Matrix:

$$\begin{bmatrix} 54 & 10 \\ 7 & 29 \end{bmatrix}$$

Classification Report (Balanced):

- Class 0 (Non-purchase): Precision **0.89**, Recall **0.84**
- Class 1 (Purchase): Precision **0.74**, Recall **0.81**
- Weighted F1-score: **0.83**

Analysis:

Applying class balancing significantly improves recall for purchasers, increasing the model's ability to correctly identify buyers. Although precision slightly decreases, the overall F1-score and accuracy improve. This demonstrates that class-weighted logistic regression provides a better trade-off between false positives and false negatives in imbalanced data.

5. K-Nearest Neighbour (KNN)

Test Set Performance:

- Accuracy: **0.91**
- Precision: **0.865**
- Recall: **0.889**
- F1-score: **0.877**

Confusion Matrix:

$$\begin{bmatrix} 59 & 5 \\ 4 & 32 \end{bmatrix}$$

Classification Report:

- Class 0: Precision **0.94**, Recall **0.92**
- Class 1: Precision **0.86**, Recall **0.89**
- Weighted F1-score: **0.91**

Analysis:

KNN outperforms logistic regression in all key metrics, achieving the highest accuracy, recall, and F1-score. The high recall indicates strong ability to detect buyers, while high precision confirms reliability of predictions. Distance-weighted KNN effectively mitigates class imbalance and captures non-linear decision boundaries.

6. K-Means Clustering and Custom Customer Types

To perform customer segmentation, **K-Means clustering** was applied using **Age** and **Estimated Salary**, without using the target variable. Features were standardized prior to clustering. The model grouped customers into two clusters, which were then mapped to **custom customer types** based on post-clustering purchase rates.

Cluster Interpretation:

- **High-Value Customer:**
Older customers with higher average salaries and significantly higher purchase probability.
- **Low-Value Customer:**
Younger customers with lower average salaries and low purchase likelihood.

Numeric cluster labels were converted into **human-readable custom data types** ("High-Value Customer" and "Low-Value Customer"), making the segmentation interpretable and suitable for business use.

7. Comparative Conclusion

Model	Accuracy	Recall (Buyers)	F1-score
Logistic Regression (Unbalanced)	0.81	0.61	0.70
Logistic Regression (Balanced)	0.83	0.81	0.77
KNN	0.91	0.89	0.88

Final Conclusion:

Class balancing significantly improves logistic regression performance by increasing recall for buyers. However, KNN achieves the best overall performance, indicating the presence of local, non-linear patterns in the data. Logistic regression remains more interpretable and stable, while KNN provides superior predictive accuracy.

Q.2

Objective

The objective of this study is to predict the sepal length of iris flowers using machine learning models and to compare the performance of KNN and SVM for this regression task.

Dataset and Feature Selection

The Iris dataset from Kaggle was used for this experiment. To improve model performance, feature engineering was applied by selecting only numerical features—sepal width, petal length, and petal width. The species feature was excluded during modeling, as it is categorical and can introduce noise in regression.

Methodology

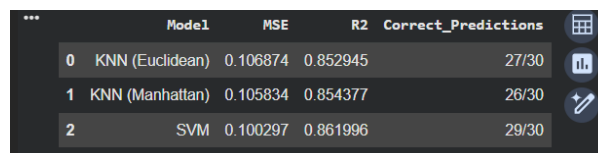
The dataset was divided into training and testing sets, and feature scaling was performed to ensure fair distance calculations. KNN models were implemented using both Euclidean and Manhattan distances, while SVM was implemented. Hyperparameters for all models were optimized using 5-fold cross-validation to avoid overfitting.

Evaluation Metrics

Model performance was evaluated using Mean Squared Error (MSE) and R^2 score. In addition, a tolerance-based accuracy metric (± 0.35 cm) was used to measure how often predictions were close enough to the true sepal length.

Results and Discussion

Among all models, SVM achieved the best performance with the lowest error, highest R^2 score, and the maximum number of correct predictions. KNN with Euclidean distance performed better than Manhattan distance, indicating that Euclidean distance is more suitable for this dataset.



	Model	MSE	R2	Correct_Predictions
0	KNN (Euclidean)	0.106874	0.852945	27/30
1	KNN (Manhattan)	0.105834	0.854377	26/30
2	SVM	0.100297	0.861996	29/30

Figure 1: Accuracy comparison.

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Accuracy (KNN Euclidean): 0.7105263157894737
Accuracy (KNN Manhattan): 0.6842105263157895
Accuracy (SVM): 0.7631578947368421
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Figure 2: Model performance metrics.

In regression problems, R^2 is a more important evaluation metric than accuracy.

Conclusion

Overall, SVM proved to be the most reliable model for predicting sepal length, while Euclidean-based KNN provided a strong and interpretable baseline.

Q.3

1. Objective

The objective of this assignment is to analyze the Titanic dataset, perform exploratory data analysis (EDA), preprocess the data, and compare multiple machine learning models to predict passenger survival.

2. Exploratory Data Analysis

EDA revealed important patterns in the data:

- The dataset is moderately imbalanced, with more non-survivors than survivors.
- Dataset has missing values in Age, Embarked and cabin column. Cabin has more NaN values.
- Female passengers had a significantly higher survival rate than males.
- Passengers in 1st class survived more frequently than those in lower classes.
- Age and Fare contained skewed distributions.
- SibSp and Parch showed skewness, motivating feature engineering.

3. Data Preprocessing and Feature Engineering

Based on EDA insights:

- Missing values were handled appropriately by removing cabin column.
- Categorical variables were encoded numerically.
- Feature scaling was applied where required.
- Class imbalance was addressed using balanced class weights in models.

4. Models Implemented

The following models were trained and evaluated:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

Each model was evaluated using accuracy and classification reports.

5. Results

Model comparison showed the following approximate accuracies:

Model	Accuracy
Logistic Regression	0.93
Random Forest	~ 0.89
SVM	~ 0.85
KNN	~ 0.74

Logistic Regression achieved the highest accuracy and was selected as the best-performing model.

6. Conclusion

This project demonstrates that effective EDA and feature engineering significantly impact model performance. Despite its simplicity, Logistic Regression outperformed more complex models, indicating strong linear relationships in the data. The final model is accurate, stable, and interpretable, making it suitable for survival prediction on the Titanic dataset.